

KSHETRA — Project Dossier

Automated Integration and Intelligent Harmonization of Multi-Source Geospatial Data for Urban Land Record Management

Team NovaX · Smart India Hackathon 2026 · Problem Statement 26013 · 21 commits, 2026-09-27 to 2026-09-30

This is the complete record of the project: the problem and the evidence for it, the design of every component we wrote, how each was evaluated, the defects we found and what they cost, and the things we deliberately refuse to claim. It is longer than a pitch needs to be, on purpose — a hackathon judge asking "how do you know that?" should find the answer here rather than a slogan.

How to read this

Two conventions run through the document.

Claims about the world are graded. [V] verified against a primary or peer-reviewed source, [S] secondary — consistently reported but not traced to the original, [U] unverified. Appendix E lists the widely-circulated statistics we decline to use.

Claims about our own system are measured, not asserted. Every number in Parts IV and VI is read out of `data/demo/metrics.json` and the other pipeline artifacts when this PDF is built. If the pipeline changes, the document changes with it; it is not possible for them to disagree.

What exists today

Component	Where	Lines	State
Harmonization engine	<code>backend/kshetra/</code>	5,181	Working, verified by harness
API with role-based redaction	<code>api/main.py</code>	385	Working, 24 routes
Operator console	<code>frontend/src/</code>	3,698	Working
Build, verification and analysis scripts	<code>scripts/</code>	2,853	Working
Deployment	<code>Dockerfile,</code> <code>render.yaml</code>	67	Builds verified; container not yet run in CI

Roughly 12,117 lines of first-party code. The engine depends on `numpy`, `scipy` and `shapely`; the serving path depends on neither, and the API image carries only `FastAPI`, `uvicorn` and `pydantic`.

One-paragraph summary

New urban orthoimagery is specified to **10 cm**. The legacy cadastral sheet it must be reconciled with has a modal positional error of **3–4 m**. Urban plots are about **11 m** across, so a legacy parcel overlaps its neighbour more than its own counterpart and naive overlay is not merely imprecise but meaningless. KSHETRA georeferences before it matches, scores candidate pairs with a model whose confidence is calibrated against observed frequency rather than invented, repairs topology without silently inventing or destroying land, resolves attribute conflicts by evidence where the quantity is measured and by legal authority where it is recorded, and — because every parcel then carries an honest uncertainty — computes where to send a survey team next and validates that recommendation against random placement. On a held-out synthetic city it recovers 95.6% of its matching decision from geometry at F1 0.9993, eliminates all 5,038 overlapping parcel pairs, and refuses to fill 69 of the 73 parcel-sized holes it finds.

Part I — The problem

Why this is hard, and how hard, with sources.

1. The structural cause

The fragmentation is neither accidental nor recent. The Survey of India was established in 1767 and conducted

revenue survey nationally until **1904**; after that date **each state became responsible for its own cadastral survey and evolved its own system** [10] [V]. Every downstream interoperability problem — divergent vocabulary, divergent scales, divergent legal primacy — descends from that one administrative decision.

1.1 Legal primacy differs between north and south

Northern states recognise the graphic map as the legal document; southern states prioritise the **Field Measurement Book** [11] [V]. In the north the polygon carries authority; in the south the *measurements* do and the polygon is a derived sketch. A single national harmonisation policy is therefore legally wrong, and a system that silently rewrites geometry is destroying the record in one half of the country while merely correcting it in the other.

This is why our conflict resolver separates measured fields from recorded ones rather than applying one rule to both (§14), and why a per-state policy object is named in the roadmap rather than a global setting.

1.2 Vocabulary differs structurally, not just lexically

South Indian survey numbers carry a recursive sub-division grammar (123/1A); northern khasra conventions vary by settlement. The record of rights is the *khatauni* in UP, the *jamabandi* in Punjab and Haryana, the *patta / chitta* in Tamil Nadu. And **"khata" denotes a revenue holding in the north but a municipal property-tax account in urban Karnataka** — a live hazard in an urban project. Schema matching must be structure-aware, not merely synonym-aware, which is why ours profiles values as well as reading column names (§13).

2. The problem, quantified

2.1 Scale of the urban gap

India has **7,933+ urban settlements** over roughly **10.2 million hectares**, with **4,912 urban local bodies** [9, 12] [V]. Only **four states** maintain structured urban land records — corroborated independently by DoLR's own NAKSHA booklet, which names Tamil Nadu, Maharashtra, Gujarat and Goa [12] [V]. About 13 million urban households live in 108,000 slums [9] [V]. Under DILRMP, **49.10%** of villages hold geo-referenced cadastral maps [13] [V].

2.2 Economic and legal consequence

Several widely-circulated "land dispute" statistics in India are recycled with drifting attribution. Only traceable figures appear here; the ones we decline to use are in Appendix D.

Claim	Source	Grade
Land market distortions erode annual GDP growth by an estimated 1.3%	Satyarthi et al., Frontiers in Sustainable Cities, 2026 [9]	[V]
ULBs generate only 32% of revenue internally, on sub-optimal tax assessment	Satyarthi et al., 2026 [9]	[V]
Land and property taxes are 0.6% of GDP in low-income economies vs 2.2% in industrialised ones	Satyarthi et al., 2026 [9]	[V]
Property disputes comprise 60–70% of Indian civil litigation	Satyarthi et al., 2026 [9]	[V] as a peer-reviewed statement
Roughly two-thirds of civil matters surveyed concerned land and property	DAKSH Access to Justice Survey 2015-16; 9,329 litigants, 305 locations, 24 states [14]	[V] as a survey of litigants, not a census
Cadastral resurvey rate of ₹56,725 per sq km	Tamil Nadu Dept. of Survey & Settlement, DILRMP review, 2024 [15]	[V]
Resurvey of a district tranche takes 2–3 years even with drone + DGPS	Tamil Nadu, 2024 [15]	[V]

3. How wrong the legacy maps actually are

Sengupta, Lemmen, Devos, Bandyopadhyay and van der Veen georeferenced **310 analogue cadastral sheets** at 1:3960 covering a 327 km² planning area in West Bengal — 258 mouzas and 26 municipal wards, mapped mostly from pre-1920s surveys and 1950s work — against GeoEye-1 pan-sharpened imagery using 10–15 ground control points per sheet and a first-order transformation [5] [V].

RMSE range	Sheets	Share
< 2.00 m	7	2.3%
2.01 – 3.00 m	64	20.6%
3.01 – 4.00 m	185	59.7%
4.01 – 5.00 m	47	15.2%
> 5.01 m	7	2.3%

The modal positional error of an Indian legacy cadastral sheet is **3–4 metres**; over 88% of sheets fall between 2 and 5 metres [5].

3.1 Transformation order is still a human judgement

The same study compared transformation orders on a sample of sheets:

Sample	1st order (m)	2nd order (m)	Reduction
1	3.668	3.177	13%
2	3.171	2.510	21%
3	3.962	1.339	66%
4	4.089	3.096	24%
5	5.010	2.889	42%

A higher-order transformation cuts error by up to two thirds on the worst sheets — but **choosing the order per sheet is manual** [5]. Automating that choice from each sheet's own residual structure is a concrete, demonstrable contribution, and it is what our escalating transform selector does (§10).

3.2 Documented failure modes [5] [V]

- **Gaps and overlaps between adjoining mouza sheets** — adjacent sheets do not fit; they overlap or leave voids.
- **Multi-sheet mouzas** with unclear division between sheets.
- **Canal double-counting** — a canal used as a shared boundary is drawn on both sheets, so the same area is counted twice.
- **River dynamics** — area lost to or accreted from a river, giving large legal-versus-digitised discrepancies.
- **Paper degradation** — shrinkage, wrinkling, folding, tearing; scanning adds further error.
- **Crude map lines** — the scanned line is often more than 4–5 pixels wide, so boundary ambiguity of decimetres to metres exists before any transformation is applied.
- **Area attribute inconsistency** — digitised parcel area does not equal the authorised area in the record of rights.

Our damage harness reproduces five of these seven directly (§17). That it is derived from a documented failure taxonomy rather than invented is the reason we consider the recovery figures meaningful.

4. What the new data will look like

Survey of India technical circular T-260/1147-Project, 10 February 2025, specifies for NAKSHA: orthorectified imagery at 5 cm GSD with **RMSE(x,y) ≤ 10 cm**; DSM/DTM at 0.5 m spacing with **RMSE(z) ≤ 15 cm**; CORS control better than 5 cm; 70%/60% overlap; a minimum of four corner and one centre GCP. The CRS is settled: **UTM on WGS 84**, vertical referenced to the Indian Vertical Datum via the Sol geoid model [6] [V].

5. The mismatch that defines the architecture

New orthoimagery: $\text{RMSE} \leq 10$ cm. Legacy sheet: modal RMSE 3–4 m. A **30–40× accuracy mismatch between the two layers the problem statement asks us to harmonise**.

Urban plots are of the order of 11 m across, so a legacy parcel displaced by 5–15 m overlaps its *neighbour* more than its own counterpart. This single comparison determines the pipeline order: georeferencing must precede matching, and any system that matches first is measuring the wrong pair.

We did not take this on faith. Matching before georeferencing gave a blocking recall of 0.58 — 42% of true correspondences never reached the model — and the model compensated by learning to join on khasra number. Reordering took blocking recall to **1.0**. The experiment is §23.2.

Part II — What the field has and has not solved

Five papers read in full, the tooling landscape, and the gap we aim at.

6. What the literature establishes

6.1 The state of the field — Chen, Nazeer, Lee & Wong [1]

A systematic review of AI in cadastre, March 2026. Deep learning has removed the manual-interpretation bottleneck in boundary extraction via CNN and Transformer segmentation; NLP methods extract structure from paper archives; deep models detect parcel change [1] [V]. It then states what remains open:

"...challenges remain, including differences in multi-temporal data processing, spatial semantic ambiguity, and the lack of large-scale, high-quality annotated data. Future research can focus on improving model generalization, advancing cross-modal data fusion, and providing recommendations for the development of a reliable and practical intelligent cadastral system." [1]

Three of those four define this project: absent annotated ground truth, cross-modal fusion, and generalisation beyond the area a method was tuned on.

6.2 Alignment under noisy supervision — Girard, Charpiat & Tarabalka [2]

The closest published statement of our core problem. A multi-resolution U-Net predicts a 2-D displacement field aligning misregistered cadastral polygons to imagery where no correct annotation exists, trained over rounds that re-correct the annotations between rounds. Alignment error fell by more than a factor of three by round two; round three added nothing [2] [V].

Their training data is built by applying known random deformations and learning to invert them — a dataset $D = \{(I, J_{\text{rand}}, f_{\text{rand}})\}$ where the deformation is generated and therefore known [2]. This is exactly our damage-harness principle, which is why we treat manufactured ground truth as established method rather than improvisation.

Two limitations we inherit. A perfect alignment score is unattainable because the reference annotations are themselves ambiguous — many buildings are outlined by a coarse polygon, so best alignment is ill-posed with multiple equally good solutions. And the model learns only *smooth* displacement fields, so adjoining buildings needing a discontinuous correction fail [2]. Our Gaussian-process error field carries the same smoothness assumption, which is why it reports an irreducible floor that no quantity of survey control removes (§15).

6.3 Parcel adjustment with SAM and ICP — Suwardhi et al. [3]

The closest operational system. UAV orthophotos at 5 cm GSD are segmented by the Segment Anything Model without manual prompting; displacement vectors come from ICP; adjustment runs hierarchically from rigid block (LS1) to scaled block (LS2) to individual parcel (LS3). Evaluated over two urban villages in Cimahi, Indonesia — Karangmekar, 81 blocks and 3,227 parcels; Baros, 96 blocks and 2,971 parcels — chosen for contrasting block structure and data quality [3] [V].

What it leaves open is the instructive part. Because no correct geometry exists for those areas either, the authors evaluate *by proxy*: they count how often cadastral polygons split the SAM-derived boundary segments, treating fewer splits as better alignment [3]. That measures internal consistency, not accuracy. It is the state of the art, and it cannot report its error in metres. It is also single-source, and does not address schema inference, CRS or unit heterogeneity, tenure semantics, calibrated confidence, or where to survey next.

6.4 Boundary revision — Fetai, Grigillo & Lisec [4]

A modified CNN detects visible land boundaries from image-based mapping and uses them to revise existing cadastral data [4] [V]. This is upstream of our pipeline and we deliberately do not rebuild it: extraction is well served, and the problem statement asks for integration of extracted features, not another extractor.

6.5 Method imported from outside the domain [8]

Our survey planner rests on sensor placement in Gaussian processes. The objective — total reduction in posterior variance over a set of observation locations — is monotone submodular, so greedy selection is within a factor $(1 - 1/e)$ of optimal and the exact problem is NP-hard [8] [V]. The result is standard in machine learning. We found no application of it to cadastral ground-control planning.

7. What the existing tools do, and do not do

Tool	What it genuinely does	What it does not do
GRASS v. clean	Topology repair: break and clean polygons from non-topological formats, remove sub-threshold areas by dissolving into the neighbour with the longest shared boundary, fuzzy snapping, vertex pruning	A human chooses the snapping and area thresholds. Wrong threshold means destroyed parcels or surviving slivers. A tool, not a decision-maker.
PostGIS Topology	Persistent topological model with shared edges and faces; prevents gaps and overlaps by construction going forward	Getting legacy dirty geometry <i>into</i> a valid topology is the hard part; common practice is to clean in GRASS first.
QGIS + GDAL/PROJ	Georeferencer with polynomial 1st–3rd order, thin-plate spline, projective and Helmert; topology checker; reprojection	All manual, per layer, per operator, with GCPs picked by hand. No cross-source reasoning.
FME / Esri Data Interoperability	Industry-standard spatial ETL; hundreds of formats; visual schema mapping	The mapping is declared by a human, not inferred. FME executes a mapping faithfully; it will not tell you that <code>KHASRA_NO</code> and <code>SY_NO</code> are the same concept, nor that they disagree for 12% of parcels. Also proprietary, at 4,912-ULB scale.
Esri Parcel Fabric	Mature cadastral editing and QA	Deed-based COGO assumptions; not built for khasra / FMB / mouza semantics or a 30–40× inter-layer accuracy mismatch.
SAM, Mask R-CNN, U-Net	Strong at extracting <i>visible</i> boundaries from imagery	Cadastral boundaries are frequently invisible — no wall, no fence, no hedge. The ceiling of the pure-vision approach.
Bhu-Naksha (the Indian incumbent)	National cadastral map software in use across many states	Manually entered per-state scale factors on shapefile import — UP ×4000, Himachal ×22 where <i>karam</i> was digitised in centimetres — and hand-written per-state RoR adapters. The incumbent national answer to unit and schema heterogeneity is a hard-coded constant. [V]

8. What is not solved

1. **No automated, evidence-weighted reconciliation across accuracy tiers.** Every tool assumes you

already know which layer is right. Nothing decides that here the 10 cm orthoimagery wins, there the legal recorded area wins, and elsewhere a human must adjudicate.

2. **Transformation-model selection is a human judgement.** Second-order beat first-order by up to 66% on the worst sheets [5], but choosing per sheet is manual.
3. **Schema matching for Indian land attributes is entirely manual.** FME and Esri require a human to draw every mapping; Bhu-Naksha's answer is bespoke per-state adapters. We found no published automated schema-matching system for Indian land-record attributes.
4. **Topology cleaning has no legal-consequence awareness.** `v.clean rmarea` will dissolve a 3 m² sliver — which may be a real, owned, taxed, litigated parcel. No tool distinguishes a digitisation artefact from a genuine tiny urban parcel, though the distinction is classifiable from area, compactness, khasra presence, RoR presence and footprint presence.
5. **Vision models find visible boundaries; cadastres contain invisible ones.** Fusing imagery, legacy geometry, textual record and utility topology to infer an invisible boundary is not addressed in the literature we found.
6. **Cross-agency entity resolution is unaddressed.** The NAKSHA progress review names "parallel, non-interoperable records" across Revenue, ULB, Development Authority and Sub-Registrar as a top friction point, lists insufficient cross-agency interoperability as a systemic void, and recommends *automated GeoAI-driven updating mechanisms* [9] [V].
7. **The state of the art is single-source and single-country.** Extending block adjustment to an n-source, schema-aware, CRS-resolving, legally-aware pipeline is what the problem statement actually asks for.
8. **No open benchmark or ground truth exists for Indian cadastral harmonisation.** Producing even a small one for a pilot ULB would be a defensible deliverable in itself.

9. Standards and the canonical target

ISO 19152 — Land Administration Domain Model. Original edition 2012; current ISO 19152-1:2024 generic conceptual model, Part 2 on land registration in development. LADM is explicitly *not* a data product specification: its purpose is "not to replace existing systems, but rather to provide a formal language for describing them." Four core packages: parties; basic administrative units carrying rights, restrictions and responsibilities; spatial units; and spatial sources and representations [16] [V].

Sengupta et al. asked in 2013 how to convert colonial maps and records into an LADM-based database, **how to document and publish the geometric quality of existing maps**, and **how to integrate more accurate data after re-survey** [17] [V]. That last question — fusing a new high-accuracy survey with an existing low-accuracy legal record without destroying the legal record — is this problem statement, posed in the literature thirteen years ago and still open.

ULPIN. A 14-character parcel identifier generated from geo-coordinates under DILRMP. The mechanism is documented; the exact character composition rule is [U] and we do not reproduce it. Our implementation is a faithful *shape*, labelled as such in code and interface.

Part III — The system

Every module we wrote, what it does, and why it is built that way.

10. Pipeline and why the order is fixed

ingest → schema inference → validate → georeference → blocking → match → assign → topology → conflict → change → uncertainty → targeting → serve

Two orderings are load-bearing and were both arrived at by being wrong first.

Georeferencing precedes matching for the reason in §5: before alignment a parcel's best geometric candidate is usually its neighbour. Measured consequence in §23.2.

Schema inference precedes validation, because you cannot check that an area column is plausible until you know which column is the area and what unit it is in. Inferring the unit from the ratio of recorded to surveyed area (§13) is only possible after geometry exists.

10.1 Runtime

Full pipeline 45.6 s on one core, 3,047 legacy parcels against 2,952 reference parcels over 52,468 candidate pairs.

Stage	Seconds
generate city	0.69
inject damage	0.3
coarse georeference	0.07
blocking	4.02
features	16.2
train matcher	0.6
assign	0.15
topology repair	2.39
fit GP	16.44
plan survey	1.76
validate plan (optimised)	0.03
validate plan (random)	0.03
schema match	0.04
build epoch t1	0.01
change detection	0.1
encroachment	0.03

Feature construction and the GP fit dominate at 16.2 s and 16.44 s. Both are embarrassingly parallel and neither has been optimised; the figure is honest rather than tuned.

11. Coordinate reference systems — `crs.py`

UTM forward and inverse, and a Helmert transformation between Everest 1830 and WGS 84, implemented from Snyder (USGS Professional Paper 1395). Written from scratch because `pyproj` would not load on the development machine (§25) — but kept, because it is now cross-validated against `pyproj` under WSL and agreement to sub-millimetre is a stronger claim than having imported the library.

Verified assertions: round-trip error < **1 mm**; central meridian easting exactly 500,000 m; agreement with `pyproj` < **1 mm** where `pyproj` is available. The last check is skipped rather than failed when it is not, so the suite runs in both environments.

12. Georeferencing — `georef/`

12.1 Coarse alignment by Hough voting

Centroid displacement candidates are accumulated into a translation accumulator and the peak taken, which survives a majority of wrong correspondences because wrong votes scatter and right ones agree. A trimmed robust similarity fit follows.

The module reports **displacement at the data centroid**, not raw t_x/t_y . With UTM northings around 3.1 million, a raw translation parameter is a number no reviewer can sanity-check; the displacement at the centroid is one they can hold against a map.

12.2 Escalating transform selection

Similarity, then affine, then second-order polynomial, accepted only when the residual structure justifies the extra degrees of freedom. This is the direct answer to §3.1 — Sengupta's 66% improvement was available but required a human to choose per sheet.

Measure	Value
RMSE before alignment	8.4529 m
RMSE after coarse alignment	3.127 m
Recovered displacement at centroid	7.37 m
Recovered rotation	-0.6163°
Recovered scale	1.001137
Inliers in the robust fit	2,133

Injected by the harness: shift (6.5, -4.2) m, rotation 0.85°, scale 1.0016. The recovered parameters are compared against those rather than against a plausibility judgement.

13. Attribute schema inference — `attributes/`

Three signals, combined and then assigned 1:1 by the Hungarian algorithm so that no two canonical fields claim the same column:

- **Lexicon match** on the column name across transliteration variants (`khasra`, `khsra`, `survey_no`, `sy_no...`).
- **Value profiling** — does the column match `n` or `n/m`? Is it unique? How many tokens per value? Do the values carry `s/o` | `w/o` | `d/o`, which marks an owner name in an Indian record?
- **Cross-field consistency** — does the candidate area column divide into surveyed area at a stable ratio?

Result on the evaluation city: **7/7 columns mapped**. Worked examples from `data/demo/schema.json`:

Canonical	Chose	Conf.	Evidence
khasra	KHSRA_NUM	0.96	name~'khsra' (0.92); 100% match <code>n</code> or <code>n/m</code> , 100% unique
owner	KHATEDAR_NM	0.9846	name~'khatedar name' (0.97); 4.9 tokens/value, 97% carry <code>s/o w/o d/o</code>
area	AREA_BIGHA	0.9167	name~'area bigha' (1.00); 100% numeric
land_use	LU_CODE	0.9932	name~'lu code' (1.00); 4 distinct values in 2842 rows
tenure	TENURE_TYP	0.9858	name~'tenure type' (0.98); 2 distinct values in 2840 rows

13.1 Inferring the unit from the geometry

The novel part. Recorded area is divided by surveyed area parcel by parcel and the modal ratio matched against known Indian units. The median came to **2500.45 m² per unit** over the evaluation set, against `bigha` at 2529.285 — inferred as **bigha** at confidence **0.9672**, with no human declaring it.

This directly addresses the Bhu-Naksha problem in §7: the incumbent national system handles unit heterogeneity with a hand-entered per-state constant. Here the constant is measured from the data.

14. Matching — `matching/`

14.1 Blocking

Candidates are ranked by polygon intersection area where it exists and by centroid proximity where it does not, so a parcel with no overlap still gets its nearest neighbours considered rather than being dropped. 52,468 candidate pairs from 3,047 sources, at blocking recall **1.0** — no true pair is lost before the model sees it.

14.2 The 29 features

Group	Features
Overlap	iou, sym_diff_norm, src_frac_in_tgt, tgt_frac_in_src
Position	centroid_dist, centroid_dist_norm, hausdorff_norm, sig_dist
Shape	area_ratio, d_compactness, d_rectangularity, d_elongation, vertex_ratio
Owner name	name_jw, name_lev, name_tok, name_present
Khasra	kh_exact, kh_parent, kh_child, kh_present
Other attributes	area_attr_ratio, land_use_match, ward_match
Context	n_candidates, iou_rank, iou_margin, is_best_iou, rev_iou_rank

The context group is what lets the model express ambiguity. A pair with IoU 0.6 means something different when it is the only candidate than when three others score 0.58, and without those features the model cannot tell the two situations apart.

Khasra features are deliberately *structural* rather than exact-match only: `kh_parent` and `kh_child` encode the recursive sub-division grammar from §1.2, so `123` matching `123/1A` is evidence of subdivision rather than a mismatch.

14.3 Model and calibration

XGBoost through the native Booster API, then **isotonic regression** fitted on held-out data so that a stated probability corresponds to an observed frequency. A weighted sum of similarity terms is not a probability and cannot be checked against anything; a calibrated one can, and that is what makes an explicit triage threshold possible instead of an arbitrary cut-off.

Measure	Value
Precision / recall / F1	0.999 / 0.9997 / 0.9993
ROC AUC	1.0
Expected calibration error	0.00009
Brier score	0.00008
TP / FP / FN / TN	2,996 / 3 / 1 / 49,468
Share of decision from geometry	95.6%

14.4 Reliability

Predicted probability against observed frequency, which is the check that the word "confidence" is being used honestly:

Bin	n	Mean predicted	Observed
0.0–0.1	49,448	0.0	0.0
0.1–0.2	3	0.1364	0.0
0.2–0.3	17	0.2	0.0588
0.3–0.4	1	0.3201	0.0
0.8–0.9	1	0.8742	0.0
0.9–1.0	2,998	0.9991	0.9993

14.5 Assignment, including splits and merges

Containment is tested *before* the Hungarian assignment, because a subdivided parcel is not a failed 1:1 match and forcing it through a 1:1 solver destroys the information. Defaults were chosen for precision over F1 — containment 0.50, split/merge coverage 0.55 — on the grounds that a wrongly asserted subdivision is a worse outcome in a

land record than an unresolved one sent to a human.

Relation	Count
One-to-one	2,791
Subdivision (one source, many targets)	148
Amalgamation (many sources, one target)	5
Unmatched source	103
Unmatched reference	77

15. Topology — `topology/cleaner.py`

Five passes: validity repair, duplicate removal, vertex snapping by union-find over cKDTree pairs, overlap resolution, and gap classification.

15.1 Principled restraint

The pass that matters most is the one that declines to act. Residual gaps are classified by area, compactness and whether anything in the record claims them. Sliver-shaped artefacts are closed; **parcel-sized holes are flagged for survey and left open**, because filling one invents land and assigns it to whoever happens to adjoin it.

Of the 73 parcels the harness deleted to create holes, the cleaner **refused to fill 69** and surfaced them for human attention.

Measure	Before	After
Invalid geometries	0	0
Overlapping parcel pairs	5,038	0
Doubly-claimed area (m ²)	18,038.6	0.0
Gaps	1,130	215
Total topology errors	6,168	215

19,277 vertices snapped; 146 slivers remain, all below the legal-consequence threshold and all reported rather than silently removed.

15.2 The area conservation ledger

Summed parcel area falls by **2.879%** across topology repair, which looks alarming until it is decomposed. Doubly-claimed land falls from 18,038.6 m² to 0.0 m², while the union footprint — ground counted once — *rises*. The summed figure falls because land recorded as owned by two parties simultaneously is now counted once.

No land was created or destroyed, and the ledger demonstrates it rather than asserting it. A system that reported only the total would look like it had lost 12,183 m² of someone's property.

16. Conflict resolution — `conflict/resolve.py`

The distinction the module is built around:

Class	Fields	Decided by
Measured	geometry, area_sqm, centroid, boundary	Evidence — inverse-variance weighting by stated accuracy
Recorded	owner, tenure, khasra, land_use, ulpin	Legal authority — the source entitled to decide

GNSS control wins on geometry; the revenue record wins on ownership, because a drone cannot observe who owns a plot. Conflating the two produces a system that will overwrite a title with a photograph.

16.1 Dominant precision, not blind averaging

Where one source is an order of magnitude better than the others, it wins outright rather than being averaged with them. From `resolutions.json`: GNSS control at $\sigma = 0.03$ m against a 1987 sheet at $\sigma = 6.0$ m and AI footprints at $\sigma = 1.2$ m, resolved to the GNSS value under rule *dominant precision*, noted "*sigma 0.03 m vs 1.2 m — combining would degrade it*". Averaging good evidence with bad evidence produces worse evidence, and a resolver that always fuses is wrong here.

16.2 Transitive inconsistency

$A \sim B$ and $B \sim C$ but $A \neq C$ cannot be resolved by any pairwise rule, and the module detects the cycle and escalates rather than picking arbitrarily. 257 cases were raised for review on the evaluation city.

17. Change detection — `change/detect.py`

Epochs are compared by spatial overlap rather than by shared identifier, because the identifier is exactly what cannot be trusted across a resurvey. DSM height delta separates a building that was extended from one that was raised.

Class	Count
New structures	31
Demolished	28
Extended (footprint grew)	70
Heightened (DSM delta, footprint stable)	59
Encroachment on public land	9 sites, 251.8 m ²

The module emits a **change dossier**, explicitly not a mutation record. A mutation is a statutory act requiring notice and hearing, and its form varies by state; producing one automatically would be both wrong and outside the problem statement. Encroachment cases are injected by the generator, which otherwise never places structures on public land — without injection there would be nothing to detect and the claim would be untested.

18. Uncertainty and survey targeting — `targeting/`

18.1 The error field

A Gaussian process is fitted to residual displacement after georeferencing, giving a posterior variance at any point — which is what "how wrong is this parcel likely to be" actually means. Fitted lengthscale **297.1 m**, which the verification suite checks is physically plausible (5–500 m) rather than a number that merely fits.

Three corrections were needed before the model was honest (§26): posterior variance must include observation noise; the kriging mean must actually be applied, since it is the correction the variance describes; and the prediction must be scaled by a factor fitted on one city and validated on another.

18.2 Planning where to survey

Candidate control locations are parcel corners. Selection is greedy submodular maximisation of total variance reduction, with the closed-form marginal gain

$$\text{gain}(c) = \Sigma_i \text{cov}(x_i, c \mid S)^2 / (\sigma^2(c \mid S) + \sigma_n^2)$$

which is within $(1 - 1/e)$ of optimal by [8]. The verification suite asserts the marginal gains are non-increasing, which is submodularity showing up as a testable property rather than a citation.

18.3 Validated, not asserted

The recommendation is then carried out: the selected control is applied, georeferencing is re-run, and achieved error is compared against the prediction *and* against random placement of the same number of points.

Measure	Value
Points recommended	19
Starting RMSE	3.127 m
Predicted after survey	1.4479 m
Achieved after survey	1.3177 m
Random placement, same budget	2.0567 m
Better than random by	35.9%

18.4 Calibration, train against held out

The prediction scale is **1.5046**, fitted on one city. Validated on a second:

Points	Train predicted	Train achieved	Held-out predicted	Held-out achieved
10	1.2052	1.7853	2.0256	1.7508
20	0.9536	1.5344	1.6567	1.4535
40	0.8623	1.293	1.522	1.3485
80	0.8121	1.2261	1.4483	1.2715

Held-out ratio median 0.8776 — the prediction errs *conservative* on unseen data, which is the direction an operational planning tool should err in. Irreducible floor 0.9009 m: per-parcel digitising noise that no quantity of control removes, reported rather than promised away.

18.5 What a recommendation looks like

Rank	Parcel	RMSE after	Gain	Reason
1	L01493	5.8271 m	-1.7043 m	highest-uncertainty region; anchors the whole sheet
2	L00647	5.0354 m	-0.7917 m	fills a gap between established control points
3	L02174	4.2905 m	-0.7449 m	fills a gap between established control points
4	L02430	3.5849 m	-0.7056 m	fills a gap between established control points
5	L00764	2.9061 m	-0.6788 m	fills a gap between established control points
6	L00843	2.5366 m	-0.3695 m	marginal refinement

19. Audit chain — `audit.py`

SHA-256 over canonical JSON plus the previous entry's hash. Every review action is appended server-side; a chain computed in the browser proves nothing, because anyone can edit it in devtools.

`verify()` distinguishes a **content** break — an entry whose payload no longer hashes to its recorded digest — from a **link** break, where an entry was removed or reordered. The two failures mean different things to an investigator, and reporting "chain invalid" for both is less useful than naming which.

`/api/audit/tamper-demo` deliberately corrupts an entry so the failure can be demonstrated live rather than described.

20. Export — `export/writers.py`

GeoJSON, Shapefile with a generated `.prj`, CSV and a PDF audit report, with a SHA-256 manifest over all of them so a recipient can verify the bundle is the one that was produced.

Two details that only appear once you actually write a shapefile: DBF field names truncate at 10 characters, so `_dbf_names()` records the mapping in a sidecar rather than letting `KHATEDAR_NM` and `KHATEDAR_NO` silently collide;

and pdf2's core fonts are latin-1, so `_PDF_FOLD` maps characters outside it to ASCII instead of crashing the export (§26).

21. API — `api/main.py`

24 routes over FastAPI. Two design decisions carry weight.

21.1 Redaction is server-side

Owner names are personal data under the DPDP Act 2023. A `public` caller never receives them — not hidden in the UI, not returned and styled away, simply absent from the response body. Doing it in the client is not a control, because the client is not trusted.

Withheld fields are *named* in a `_redacted` marker rather than silently dropped, so a consumer can tell a withheld field from a missing one. A silently truncated record is worse than a marked one.

Role	Sees owner	Sees tenure	May act
Public	no	no	—
Surveyor	no	yes	raise for survey
Revenue clerk	yes	yes	escalate, raise for survey
Tehsildar	yes	yes	accept, reject, escalate, survey

21.2 OGC API — Features

`/ogc/collections/{id}/items` returns `application/geo+json` with `numberMatched` and `numberReturned`, so any conformant GIS client can consume the harmonized output directly without an adapter. Redaction applies there too — a standards-compliant endpoint is not a way around the access control.

22. Operator console — `frontend/src/`

3,698 lines of TypeScript, React and CSS. MapLibre GL JS v4 with three Esri basemaps, `fill-extrusion` for surveyed building heights, and an alignment animation that interpolates between the raw and georeferenced layers so the correction is something a judge watches happen rather than reads about.

Component	Lines	Role
<code>MapView.tsx</code>	545	Map, layers, basemaps, 3D, alignment animation
<code>Dock.tsx</code> + <code>DockExtra.tsx</code>	828	Review queue, metrics, validation, survey plan, export
<code>Panels.tsx</code>	251	Layer tree, parcel inspector
<code>Palette.tsx</code>	236	Ctrl+K command palette, role-aware
<code>DemoMode.tsx</code>	230	Eight-beat guided walkthrough, figures read from artifacts
<code>SignIn.tsx</code>	176	Credential sign-in; role is a property of the account
<code>Guards.tsx</code>	73	Error boundary, offline banner
<code>store.ts</code>	337	State, role table mirroring the server's, redaction

The role table in `store.ts` mirrors `VISIBLE` and `MAY_ACT` in `api/main.py` deliberately: the UI must not offer an action the server would refuse, nor display a field the server would withhold. The server remains the enforcement point; the client copy exists so the interface does not lie about what is available.

Sign-in reports the same error for an unknown username and a wrong password, because saying which half failed tells an attacker which usernames exist.

Part IV — Evaluation

How we know, rather than what we hope.

23. Methodology: manufacturing the ground truth

Both [2] and [3] are limited by the same thing — no correct geometry exists to score against, so one reports relative improvement and the other a proxy. We generate a clean synthetic cadastre, keep it as the key, damage a copy through twelve stages whose parameters are recorded, and score recovery against the key.

This is not a workaround. It is the method [2] uses, stated explicitly: build the dataset by applying known deformations and learn to invert them. The difference is that our deformations are drawn from a *documented* failure taxonomy (§3.2) rather than chosen for convenience.

23.1 The damage model

Stage	Parameter	Real-world cause
Similarity misregistration	shift (6.5, -4.2) m, rotation 0.85°, scale 1.0016	sheet georeferenced from too few, poorly spread control points
Smooth non-linear warp	amplitude 2.4 m over 9 control points	paper shrinkage, scanner distortion, map-sheet join
Vertex jitter	0.35 m per vertex	manual digitisation from a paper sheet
Vertex decimation	10% of vertices dropped	generalisation during digitisation
Sliver injection	6% of parcels shrunk by 0.45 m	adjoining sheets digitised independently, leaving voids
Overlap injection	5% of parcels grown by 0.55 m	shared boundaries double-counted between sheets
Self-intersection	1.2% bow-tied, 4% given duplicate vertices	careless digitising, unclosed rings
Attribute schema drift	all 7 columns renamed and reordered	every department names its columns differently
Owner-name corruption	14% transliteration typos, 5% missing	transliteration variance across romanisations
Area unit drift	m ² recorded as bigha	records kept in bigha and biswa, not metric
Khasra renumbering	45% of blocks renumbered	resurvey reissues parcel numbers
Mutation	4% split, 3% merged, 2.5% deleted, 2% spurious	subdivision, amalgamation and record loss since the sheet was drawn

23.2 Two experiments that changed the design

Pipeline order. Matching before georeferencing gave blocking recall 0.58. 42% of true correspondences never reached the model, and it compensated by learning attribute joins. After reordering: **1.0**. The architecture in §10 is the result of that measurement, not of intuition.

Leakage. The first matcher scored F1 0.9955 with khasra and owner supplying 96% of the decision and IoU 0.24%. It had learned to join on an ID, which fails in any jurisdiction that renumbers on resurvey — that is to say, in the exact situation the problem statement describes. The harness now renumbers 45% of blocks and genuinely changes 12% of owners, so the shortcut no longer exists.

24. The verification suite — `scripts/verify_all.py`

Eight stages, roughly 25 assertions, run end to end. These are the ones that would fail loudly if a change broke a claim this document makes:

Assertion	Guards against
round-trip < 1 mm	CRS implementation drift
central meridian == 500000 m exactly	UTM false easting error
agrees with pyproj < 1 mm	our hand-rolled CRS being subtly wrong
no overlaps in ground truth	a dirty key, which would make every later figure meaningless
damage actually applied	scoring recovery on an undamaged copy
coarse alignment reduces error > 40%	georeferencing silently regressing
blocking recall >= 0.98	the 0.58 failure recurring unnoticed
F1 >= 0.90	matcher regression
calibration ECE <= 0.02	confidence becoming decorative
geometry drives the model >= 70% gain	the leakage trap returning
exact set match >= 0.85	assignment degrading on splits and merges
spurious parcels rejected >= 0.95	accepting records that should not exist
topology errors reduced >= 90%	cleaner regression
refuses to fill parcel-sized holes	the restraint in §15.1 being optimised away
lengthscale plausible (5-500 m)	a GP that fits but means nothing
marginal gains non-increasing	submodularity being violated by an implementation bug
optimised placement beats random	the central targeting claim

The geometry-share and hole-refusal assertions are there because both properties are the kind a well-meaning optimisation would remove. A model that is allowed to use khasra numbers will score better; a cleaner that fills every hole will report fewer errors. Both would be worse systems, and a test is the only thing that prevents the improvement.

25. Results

Trained on one synthetic city, evaluated on a second with a different layout and a different damage profile.

Stage	Measure	Result
Georeferencing	positional RMSE	8.4529 m → 3.127 m
Schema inference	columns mapped	7/7
Schema inference	area unit from geometry	bigha at 0.9672
Blocking	true pairs retained	1.0
Matching	precision / recall / F1	0.999 / 0.9997 / 0.9993
Matching	ROC AUC	1.0
Matching	decision from geometry	95.6%
Confidence	expected calibration error	0.00009
Assignment	1:1 / subdivision / amalgamation	2,791 / 148 / 5
Topology	total errors	6,168 → 215
Topology	doubly-claimed land	18,038.6 → 0.0 m ²
Restraint	parcel-sized gaps refused	69 of 73

Change	new / demolished / extended / heightened	31 / 28 / 70 / 59
Encroachment	sites on public land	9, 251.8 m ²
Targeting	predicted / achieved / random	1.4479 / 1.3177 / 2.0567 m

25.1 Where the matching decision comes from

Top ten features by total gain. This table is the evidence for the headline claim that the system matches on *shape and position*, not on an identifier that a resurvey will change:

Feature	Gain
src_frac_in_tgt	0.46506
centroid_dist	0.39963
iou	0.06393
kh_present	0.0174
sym_diff_norm	0.01576
kh_child	0.01027
is_best_iou	0.00436
vertex_ratio	0.00265
iou_rank	0.00228
d_compactness	0.00223

The three leading features — containment, centroid distance and IoU — are purely geometric and account for 92.9% of total gain between them. `kh_present` ranks fourth at 0.0174, and note what it encodes: not which khasra number, but *whether one is present at all* — a missingness signal, not a join key.

Part V — Engineering record

What broke, what it cost, and how the environment was made to work at all.

26. The platform problem

Windows **Smart App Control** blocked the scientific Python stack on the development machine: `pyproj`, `scikit-learn` and `LightGBM` at every version tried, and `numpy` 2.0.2 while permitting 2.4.6. The blocking is per-binary and not documented in advance, so it presents as an unexplained `ImportError` on a package that installed cleanly.

Two consequences shaped the codebase, and both turned out to be improvements.

The CRS engine was written from scratch (§11). Now cross-checked against `pyproj` to sub-millimetre, which is a stronger statement than importing it would have been.

The engine degrades gracefully when an optional dependency is absent. Metrics are hand-implemented rather than imported from `scikit-learn`; the `pyproj` cross-check skips rather than fails. This is why the project survived when, on 26 September, Smart App Control was auto-promoted from evaluation to enforced and stopped everything at once.

The fix was WSL2 — where, without `sudo`, `ensurepip` is unavailable, so the environment is built with `python3 -m venv --without-pip` followed by `get-pip.py`, and Node is installed from a tarball into `~/.local`. Both are scripted in `wsl_bootstrap.sh`.

Smart App Control also blocks Git Bash's bundled `ssh`, which surfaces as `cannot spawn ssh: Function not implemented` on push. The repository sets `core.sshCommand` to the Windows OpenSSH binary to work around it.

27. Defects found, and what each cost

Listed because the interesting part of a project is rarely what worked first time. Several of these were found by using the system rather than by reading it, which is an argument for building something operable early.

Defect	What was wrong	Resolution
The leakage trap	First matcher scored F1 0.9955 with khasra number and owner name supplying 96% of the decision and IoU 0.24%. It had learned to join on an ID, which fails in any jurisdiction that rennumbers on resurvey.	Harness now rennumbers 45% of blocks and genuinely changes 12% of owners. <code>verify_all.py</code> asserts geometry supplies $\geq 70\%$ of total gain. It supplies 95.6%.
Blocking before georeferencing	Candidate generation ran on raw coordinates, so blocking recall was 0.58 -- 42% of true correspondences never reached the model at all, and the model compensated by leaning on attributes.	Pipeline reordered so georeferencing precedes blocking. Recall 1.0.
Gaussian-process over-optimism	Predicted post-survey RMSE 0.61 m against 1.23 m achieved. Three causes: posterior variance quoted without observation noise, no kriging-mean correction, and an uncalibrated prediction scale.	All three fixed, and the prediction scale is now fitted on one city and validated on a held-out one. Reported honestly: the 'better than random' margin fell from 51.8% to 35.9%, because a better interpolator also rescues random placement. The old figure was partly an artifact.
fitBounds over the first 400 features	Features are block-ordered, so the first 400 describe a 170 m strip. The opening camera zoomed into a sliver of the ward.	Bounds computed over the full collection.
Layer-visibility race	<code>boot()</code> added map layers after the sync effect had last run, so the uncertainty field rendered as a yellow haze over everything.	A ready flag added to the effect dependencies.
Percentage widths on inline elements	Feature-importance bars were spans; a percentage width on an inline element is ignored, so every bar rendered at zero pixels.	Both track and fill declared <code>display:block</code> .
GeometryCollection from gap filling	Filling a gap occasionally produced a GeometryCollection, which crashed the shapefile writer.	<code>_largest_polygon()</code> added at the export boundary.
Sign-in overlay inside the stage	The toolbar rendered 'Tehsildar / FULL RECORD' before anyone had signed in -- a role indicator showing a role nobody held.	Overlay moved to <code>position:fixed</code> with its own stacking context.
Esri canvas basemaps stop at zoom 16	Switching basemap while pitched (z16.8) requested tiles that do not exist, and the map went blank.	Per-basemap <code>maxzoom</code> declared on each raster source.
Command-palette PII leak	Owner names were filtered out of palette <code>*search*</code> for the public role but still printed as result subtitles. Found by testing as public rather than by reading the code.	<code>redact()</code> applied at the point of render, not at the point of query.
Map controls behind the dock	A fixed 52 px offset put the basemap and 3D controls underneath the dock whenever it was open.	Offset derived from dock state.

autocrlf rewrote the shell scripts	Git's line-ending conversion broke every path in <code>.sync.sh</code> under <code>bash</code> .	<code>.gitattributes</code> pins LF for <code>.sh</code> , <code>.py</code> , <code>Dockerfile</code> and <code>YAML</code> .
fpdf2 core fonts are latin-1	The audit report PDF rejected an em-dash and crashed the export.	<code>_PDF_FOLD</code> maps non-latin-1 characters to ASCII equivalents.

27.1 The one worth dwelling on

The Gaussian-process over-optimism is the defect we are most inclined to report, because fixing it made a headline number *worse*. "Better than random" fell from 51.8% to 35.9% — not because the planner got worse, but because a better interpolator also rescues random placement, so the old margin was partly an artifact of the bug. A system whose reported advantage goes down when its model is corrected is a system whose numbers were previously flattering it.

28. Development toolchain

Script	Lines	Purpose
<code>verify_all.py</code>	336	End-to-end verification; ~25 assertions across 8 stages
<code>build_demo.py</code>	657	Runs the pipeline, writes the WGS84 artifacts the UI reads
<code>calibrate_targeting.py</code>	152	Fits the prediction scale on one city, validates on a held-out one
<code>fetch_data.py</code>	216	Microsoft and OSM sources, resumable, with Overpass mirror rotation
<code>check_gp_calibration.py</code>	108	Isolates the GP prediction-versus-achieved question
<code>check_api.py</code>	137	Exercises every route, including redaction by role
<code>serve.py</code>	53	No-cache static server; a stale <code>index.html</code> cost real debugging time

29. Deployment

A single Docker image serves the built UI and the API from one origin. Two services would be cheaper to reason about, but then the page and the API sit on different origins and the redaction and the audit chain need CORS to agree with them; one container keeps the security story intact.

Stage	Base	Does
1	<code>node:20-alpine</code>	<code>npm install</code> , copy pipeline artifacts into <code>public/data</code> , <code>tsc -b</code> && vite build
2	<code>python:3.12-slim</code>	Install FastAPI, uvicorn, pydantic; copy backend, api, data and the built dist; run <code>uvicorn</code>

The API imports `kshetra.audit`, which is `stdlib` `hashlib` and `json` — nothing in the request path needs `numpy`, `scipy`, `shapely` or `xgboost`. Those build the artifacts offline, so `requirements-api.txt` carries the serving dependencies only and the image is roughly 200 MB rather than 900 MB.

Pipeline artifacts are baked into the image rather than fetched at boot, so the page renders while a cold instance is still waking. The UI mount is conditional on `dist/` existing, so local development with Vite on its own port is unaffected.

Known deployment limitations: on a free instance the service spins down after 15 minutes idle with a roughly 50-second cold start, and the audit chain writes to ephemeral disk, so it persists within a session and resets on redeploy. Neither is a design property; both are the hosting tier.

Part VI — Limits and what comes next

Stated plainly, because a reviewer will find them anyway.

30. Limitations

- The cadastral parcel layer is **synthetic**, derived from OpenStreetMap block structure and labelled as such throughout the product and its exports. No bulk Indian urban cadastre is publicly available. Building footprints, the road network and the terrain are real.
- Encroachment cases are **deliberately injected**: the generator never places structures on public land, so without injection there would be nothing to detect and the claim would be untested.
- The survey-plan prediction is **calibrated, not exact**. A scale factor measured on one city transfers to a held-out city within roughly 12%, and errs conservative.
- Per-parcel digitising noise is **irreducible** by survey control. The system reports that floor (0.9009 m here) rather than promising accuracy below it — a limitation shared with [2], whose displacement model is likewise smooth.
- The system detects and evidences change. It does **not** produce a mutation record, which is a statutory act requiring notice and hearing and whose form varies by state.
- Redaction is enforced server-side by the API. In the static demonstration build every artifact is delivered to the browser, so redaction *there* is a display control only. The distinction is documented in the repository and stated here rather than glossed.
- Evaluation is on two synthetic cities. Generalisation to a third, real ward is untested, and we would not claim it from these numbers.
- The Docker image has been reasoned about and its build stages reproduced by hand, but the container itself has not yet been run in CI.

31. Open questions we would take further

- **Sliver classification with legal consequence.** Distinguishing a digitisation artefact from a genuine tiny urban parcel is classifiable from area, compactness, khasra presence, RoR presence and footprint presence. We currently classify on geometry alone and flag the rest.
- **Invisible boundaries.** Fusing imagery, legacy geometry, textual record and utility topology to infer a boundary with no physical trace is the ceiling of the vision-only approach and unaddressed in the literature we found.
- **North/south legal regimes.** Preserving measurement authority where the Field Measurement Book is primary needs a per-state policy object rather than a global setting.
- **Automatic transformation-order selection at scale.** We escalate; Sengupta's data suggests per-sheet selection is worth up to 66%, and quantifying our selector against their 310-sheet distribution would be a real result.
- **An Indian benchmark.** A small, released, ground-truthed harmonisation benchmark for one pilot ULB would let this field measure itself.

Appendix A — module inventory

Tracked first-party Python over 100 lines. Frontend inventory is in §22.

File	Lines
backend/kshetra/synth/corruption.py	464
api/main.py	385
backend/kshetra/topology/cleaner.py	384

backend/kshetra/attributes/schema_match.py	348
backend/kshetra/export/writers.py	325
backend/kshetra/synth/generator.py	306
backend/kshetra/change/detect.py	299
backend/kshetra/conflict/resolve.py	299
backend/kshetra/targeting/uncertainty.py	278
backend/kshetra/targeting/planner.py	275
backend/kshetra/georef/transform.py	254
backend/kshetra/matching/assign.py	250
backend/kshetra/crs.py	231
backend/kshetra/matching/features.py	230
backend/kshetra/matching/model.py	200
backend/kshetra/attributes/text.py	191
backend/kshetra/georef/coarse.py	186
backend/kshetra/audit.py	174
backend/kshetra/geometry.py	170
backend/kshetra/layers.py	154
backend/kshetra/evaluation/metrics.py	148

Appendix B — commit history

Commit	Date	Subject
fc84d56	2026-09-30	Pin LF on Dockerfile and YAML
b67e296	2026-09-30	Deploy: single Docker image serving the UI and the API on one origin
67af6d5	2026-09-30	Expand the research summary from 4 pages to 9
c9cb84a	2026-09-30	Research summary PDF with paper-backed positioning
0c252f1	2026-09-30	Credential sign-in; 3D map, basemap switcher, bolder cartography
764682e	2026-09-29	Slide 1 and 3 mockups in the team's existing SIH idiom
4e013cc	2026-09-29	Role sign-in, KSHETRA mark, command palette
f38ca3a	2026-09-29	Round the figures in the demo script
7999a34	2026-09-28	Guided demo mode; no-cache server
0027ad7	2026-09-28	Light theme; basemap degrades gracefully; error boundaries
8d30992	2026-09-28	Calibrate the survey-plan prediction; correct with the GP, not a spline
394f0e3	2026-09-28	Add Export dock tab and a standalone launcher
fafba65	2026-09-28	Tier 2: server-side audit chain, export writers, FastAPI with RBAC
7d76d4c	2026-09-28	Pin LF endings for shell scripts; drop one-shot patch scripts
895c375	2026-09-28	Rename to KSHETRA; add schema matching, conflict resolution, change detection
252a9a9	2026-09-28	Working MVP: engine verified, demo artifacts, mission-control UI
d8e7fd7	2026-09-27	Add PROMPT.md -- single pasteable build prompt

5d10eb0	2026-09-27	Deepen build prompt: conflict taxonomy, interaction spec, failure modes
f719fd7	2026-09-27	Add styled handoff page with print stylesheet
ca532b7	2026-09-27	Add HANDOFF.md for device migration
2a53916	2026-09-27	BhoomiSetu: engine, targeting module, real data, build plan

Appendix C — references

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Appendix D — claims we deliberately do not make

Several figures circulate widely in Indian land-administration discussion without traceable provenance. We exclude them rather than risk being unable to defend them:

- **"Land disputes take about 20 years to resolve."** Attributed to NITI Aayog across many secondary sources; we could not open the original. [S]
- **"324 years to clear the case backlog."** Source not located. [U]
- **"25% of Supreme Court decided cases involve land disputes."** Widely repeated, original study not

found. [U]

- **Any figure for man-hours or rupee cost of manual GIS harmonisation** specifically, as distinct from survey or digitisation cost. We searched and found nothing credible in the public domain and do not invent one. Where a cost proxy is needed we use the ₹56,725 per sq km resurvey rate [15], which is official.
- **"Indian Geodetic Datum 2023."** This does not exist. The National Geospatial Policy 2022 [7] commits to *redefining* the national geodetic framework; the Indian Vertical Datum is real and is referenced in [6].
- **The exact 14-character composition of ULPIN**, and the contents of the NAKSHA SDMS schema, whose SOP is available only as a scan without a text layer. [U]

Sources [1]–[4] and [8] were read in the original. Sources [5]–[7] and [9]–[17] are drawn from a sourced research dossier compiled for this project; figures attributed to them carry their grade, and any figure intended for publication should be re-checked against the primary document.